CEM AKSU

Embedded Systems | Electrical and Electronics Engineering Student
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SUMMARY

Senior Electrical and Electronics Engineering student with hands-on STM32 experience from an R&D internship, covering peripheral configuration in STM32CubeMX, firmware development in C with the STM32 HAL, and on-target debugging. Also builds Python tools for data processing and simulation, including a 20-year portfolio backtesting engine. Seeking an embedded systems internship or junior role.

EDUCATION

Yaşar University İzmir, Türkiye
B.Sc. in Electrical and Electronics Engineering *4th-year student*

EXPERIENCE

ENKO Control Systems Inc. İzmir, Türkiye
R&D Intern, Embedded Systems *Jul 2026 – Aug 2026*

- Configured STM32 microcontrollers (STM32F407 and STM32F429) in STM32CubeMX, including clock tree, GPIO inputs and outputs, SysTick and NVIC settings.
- Developed and debugged firmware in C using the STM32 HAL in STM32CubeIDE and flashed boards with STM32CubeProgrammer.
- Co-developed, in a team of three interns, a reusable non-blocking LED driver module in C based on a state machine and HAL_GetTick timing, running several outputs with independent on and off periods.
- Implemented a SysTick-based 1 ms time-sliced task loop to schedule alternating tasks without an RTOS.
- Used board user manuals and MCU reference documentation to map pins and peripherals, and worked alongside mechanical engineering interns in a multidisciplinary R&D team.

PROJECTS

BIST Quant: Dividend-Growth Portfolio Backtesting Engine 2026
Python, pandas, NumPy, Streamlit, Plotly, yfinance, PyInstaller

- Built a backtesting engine that simulates a long-term dividend portfolio on Borsa Istanbul over 20 years (2006 to 2026) with annual rebalancing and monthly contributions, using only trailing data to avoid look-ahead bias.
- Designed a Markov-state trust model that removes stocks after a skipped dividend and re-admits them after a configurable trust period, with weights based on trailing 10-year CAGR.
- Reported XIRR, Sharpe ratio, Sortino ratio and maximum drawdown in USD terms, and packaged the app as a Windows executable across four iterative versions.

TEFAS Quant: Fund Analysis and Financial Independence Terminal 2026
Python, Streamlit, pandas, Plotly, pyTEFAS

- Developed a dashboard that pulls Turkish mutual fund data from TEFAS, converts returns to USD and ranks funds by CAGR.
- Modeled financial independence (FIRE) targets by withdrawal rate and inflation, comparing conservative and aggressive scenarios.

Asynchronous Algorithmic Trading Bot 2026
Python, asyncio, pandas, NumPy

- Structured an event loop as separate data, strategy, risk and execution modules, generating EMA, RSI and ADX signals on each new bar.
- Added volatility-based position sizing and a trailing stop, with shared state protected by asyncio locks.

TECHNICAL SKILLS

Programming: C (Embedded C), Python
Embedded: STM32 (F4 series), STM32 HAL, STM32CubeIDE, STM32CubeMX, STM32CubeProgrammer, GPIO, SysTick, clock configuration
Simulation and Design: MATLAB, Proteus, LTspice, Logisim
Python Libraries: pandas, NumPy, Plotly, Streamlit, PyInstaller
Tools: Git, GitHub
Languages: Turkish (Native), English (Advanced)

CERTIFICATIONS

Signal Processing Onramp, MathWorks November 2025